

WELL COMPLETION REPORT (SYNTHETIC)

OIL-DRK-003 - Dirok-003

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000032
Field / District	Dirok / Dibrugarh
Block	OALP-III-08
Basin	Assam-Arakan
Formation	Kopili (KOP)
Spud / Completion	2009-07-31 / 2009-11-19
Measured / TVD	3423 m / 3409 m
Trajectory	Vertical; inc 5.4 deg; az 242.1 deg
Formation Interval	1659-2625 m
Lithology	Calcareous shale and siltstone
Pore Pressure Reference	3549 psi
Fracture Gradient	15.71 ppg
Mud System	KCl Polymer WBM
NWIS Well Risk	Moderate (51.6/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	855	Surface
9 5/8 in casing	1985	Intermediate
7 in liner/casing	3012	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2009-08-12 523 m	Kick	High	Drilling break followed by positive flow check. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
2009-08-17 684 m	Cementing Issue	High	Cement returns were below planned percentage. Mitigation: Review loss history before cementing equivalent interval. Lesson: Plan conservative displacement rate in weak formations.
2009-10-09 2258 m	Torque Spike	High	High torque developed while WOB remained elevated. Mitigation: Reduce WOB/RPM and circulate before resuming normal parameters. Lesson: Use offset torque trend to set adaptive operating envelope.
2009-10-21 2611 m	Cementing Issue	High	Displacement pressure became unstable. Mitigation: Control displacement rate and reconcile pumped/returned volumes. Lesson: Plan conservative displacement rate in weak formations.
2009-11-09 3131 m	Mud Loss	High	Partial returns observed at shale shaker. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Track flow balance and pit volume trend before losses become severe.
2009-08-18 703 m	Mud Loss	Medium	Intermittent circulation losses developed while drilling ahead. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
2009-08-24 899 m	NPT	Medium	No depth progress occurred during the recorded interval. Mitigation: Verify equipment health before resuming drilling. Lesson: Consistent NPT coding enables meaningful cross-well comparison.

Completion Summary

The synthetic well reached 3423 m MD in the Kopili interval. The NWIS knowledge record links 12 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.